

# G M Tasnim Alam

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## SUMMARY

Ph.D. candidate in Computer Science specializing in **protocol security and automated vulnerability discovery**, currently focused on 5G cellular networks. Discovered **18 CVEs** (one critical, CVSS 9.8) in open-source 5G core implementations through fuzzing-based protocol testing. Co-author of an ACM WiSec 2025 publication. Industry experience as a Software Engineering Intern at **Apple** and Senior Software Engineer at **Agoda** (Booking Holdings).

## EDUCATION

**Stony Brook University (SUNY)**  
*Ph.D. in Computer Science, Reasoners Lab (Expected: 2028)*  

- Research Focus:** Protocol security and automated vulnerability discovery in 5G cellular networks via stateful fuzzing.
- Coursework:** Systems Security, Network Security, Algorithms, Big Data, Data Science, Visualization.

Aug 2023 – Present  
Stony Brook, NY

## RESEARCH EXPERIENCE

**Stateful Fuzzing of 5G Core Networks**  
*Ph.D. Research, Reasoners Lab, Stony Brook University*  

- Designed and built a black-box adaptive fuzzer for 5G Core networks that mutates **NGAP** control-plane messages and exercises stateful multi-step scenarios across AMF/SMF/UPF components.
- Developed automated bug triage that classifies anomalies (crash, hang, protocol-state violation, response divergence) by observing system behavior and downstream responses.
- Discovered **18 CVEs** in open-source 5G core stacks (OpenAirInterface, Linux Foundation projects), including a **critical CVSS 9.8**, four High (CVSS 7.5), and three Medium-severity issues.
- Disclosed all findings to upstream maintainers under coordinated disclosure.

Aug 2023 – Present  
Stony Brook, NY

**eSIM M2M Remote Provisioning Protocol Analysis**  
*Co-author, ACM WiSec 2025*  

- Designed and conducted the experimental evaluation supporting the paper's theoretical claims on the performance-consistency trade-off of the eSIM M2M remote provisioning protocol.
- Built the testbed, automated measurement pipeline, and analysis scripts used to validate the protocol model end-to-end.

2024 – 2025  
Stony Brook, NY

## INDUSTRY EXPERIENCE

**Apple Inc.**  
*Software Engineering Intern, Accessibility Organization*  

- Owned a full iOS feature project end-to-end (design, implementation, testing) targeting users with limited mobility, working in Swift across the Accessibility framework stack.
- Delivered a production-ready implementation that shipped to internal beta and was validated through cross-team design, engineering, and accessibility reviews.

May 2025 – Aug 2025  
Cupertino, CA

**Agoda (Booking Holdings)**  
*Senior Software Engineer, iOS*  

- Designed and shipped a multi-product shopping-cart and checkout module for the Agoda iOS app using modular architecture patterns, contributing to a measurable lift in checkout conversion.
- Built and maintained iOS booking workflow components integrating third-party APIs and analytics, improving reliability and reducing friction across the purchase funnel.
- Led migration of analytics components from a C#-based shared layer to a native Swift implementation on iOS, improving event-data correctness and on-device processing efficiency.

Jan 2022 – June 2023  
Bangkok, Thailand

## PUBLICATIONS

**Performance and Consistency Trade-off of the eSIM M2M Remote Provisioning Protocol.** ACM WiSec 2025. [ACM DL]

## CVEs / VULNERABILITY DISCLOSURES

**18 CVEs** in open-source 5G core implementations (OpenAirInterface Alliance, Linux Foundation projects), discovered through stateful protocol fuzzing research:

**Critical (CVSS 9.8):** CVE-2026-30075    **High (CVSS 7.5):** CVE-2026-30077, CVE-2026-30078, CVE-2026-30079, CVE-2026-30080    **Medium:** CVE-2026-5360, CVE-2026-4531, CVE-2026-5661, CVE-2026-8783, CVE-2026-8782, CVE-2026-8781, CVE-2026-8780, CVE-2026-8779, CVE-2026-8349, CVE-2026-9301, CVE-2026-9299, CVE-2026-9300, CVE-2026-9298

## SELECTED PROJECTS

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### **CI/CD Pipeline Security Monitoring** | *osquery, eBPF, Kafka, Logstash, Elasticsearch, Docker*

- Built a DevSecOps telemetry pipeline for GitHub Actions runners, collecting host- and workflow-level events via osquery and eBPF, streaming through Kafka, and indexing in Elasticsearch for detection, auditing, and incident triage.

### **Malicious Repository Detection at Scale** | *Big Data, Python*

- Implemented scalable data pipelines and detection heuristics to flag potentially malicious open-source repositories using big-data techniques over large code-hosting datasets.

### **NYC Motor Vehicle Collision Analytics** | *Python, D3.js*

- Built a data cleaning, feature engineering, and interactive D3.js visualization pipeline over NYC collision datasets to surface trends, hotspots, and risk patterns.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, Swift, Java

**Security & Fuzzing:** Stateful protocol fuzzing, NGAP/5G NAS, vulnerability triage, coordinated disclosure, eBPF, osquery

**Networking & Protocols:** 5G Core (AMF/SMF/UPF), eSIM/M2M provisioning, REST APIs, TCP/IP

**Systems & Infra:** Linux, Docker, Kafka, Elasticsearch, Logstash, Git, GitHub Actions, CI/CD

**Platforms:** iOS / Swift, microservices

## TEACHING, SERVICE & AWARDS

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**Teaching Assistant**, Stony Brook University: CSE 541 Logic in Computer Science, CSE 381 Game Programming, CSE 416 Software Engineering.

**Sub-reviewer**, IEEE Symposium on Security and Privacy (S&P), 2025.

**Cyber Powder Fellow**, University of Utah, 2025.